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Sound analysis/synthesis tools for music applications

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README
License

sms-tools

Sound analysis/synthesis tools for music applications written in python (with a bit of C) plus complementary lecture materials.

How to use

In order to use these tools you have to install python (2.7 or greater) and the following modules: ipython, numpy, matplotlib, scipy, and cython.

In Ubuntu (which we strongly recommend) in order to install all these modules it is as simple as typing in the Terminal:

```
$ sudo apt-get install python-dev ipython python-numpy python-matplotlib python-scipy cython
```

In OSX (which we do not support but that should work) you install these modules by typing in the Terminal:

```
$ pip install ipython numpy matplotlib scipy cython
```

then, for using the tools, after downloading the whole package, you need to compile some C functions. For that you should go to the directory `software/models/utilFunctions_C` and type:

```
$ python compileModule.py build_ext --inplace
```

The basic sound analysis/synthesis functions, or models, are in the directory `software/models` and there is a graphical interface and individual example functions in `software/models_interface`. To execute the models GUI you have to go to the directory `software/models_interface` and type:

```
$ python models_GUI.py
```

To execute the transformations GUI that calls various sound transformation functions go to the directory `software/transformations_interface` and type:

```
$ python transformations_GUI.py
```

To modify the existing code, or to create your own using some of the functions, we recommend to use the `workspace` directory. Typically you would copy a file from `software/models_interface` or from `software/transformations_interface` to that directory, modify the code, and execute it from there (you will have to change some of the paths inside the files).

Content

All the code is in the `software` directory, with subdirectories for the models, the transformations, and the interfaces. The lecture materials are in the `lectures` directory,

the assignments related to the lectures in the `assignments` directory, and the sounds used for the examples and coming from `http://freesound.org` are in the `sounds` directory.

License

All the software is distributed with the Affero GPL license (http://www.gnu.org/licenses/agpl-3.0.en.html), the lecture slides are distributed under Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0) license (http://creativecommons.org/licenses/by-nc-sa/4.0/) and the sounds in this repository are released under Creative Commons Attribution 4.0 (CC BY 4.0) license

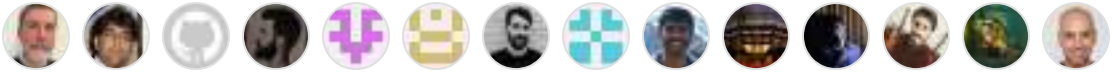
Releases

No releases published

Packages 0

No packages published

Contributors 33



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Languages



Python 82.9% **C** 14.8% **Cython** 2.3%

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